

technical notes

- Data format. It is a json file which is a list of lists, in python: ['mass pm 1.0', 'mass pm 2.5', 'mass pm 4.0', 'mass pm 10', 'number pm 0.5', 'number pm 1.0', 'number pm 2.5', 'number pm 4.0', 'number pm 10', 'typical size']
- All tests are done with the filter appliance in the middle of the room, unless otherwise stated (just one in the beginning was done on a table at the end of the room to see what effect moving the machine might have)
- I acknowledge the possibility of bias, mostly unconscious, however someone has to actually do the measurements and no one else seems to be doing any, or has measured the lasko fan at lower speeds/noise levels. I offer to mail the equipment to an academic in the field who would like to do similar testing, or someone else the community could trust more.
 - Ideally someone in montreal so the filters fans and everything can be sent.
- It's Important to note that while the BQF may be considered equivalent to X many fans, that does not mean that subbing it in place of 1 fan with only 1 filter set will give X times the cadr. Consider pictures of pressure flow curves for fan and filters. Where they intersect does not increase by X times just because you have more fans, or you multiply the y axis by A to indicate increased flow at any given pressure of the fan combination (2x for instance). However, if you balance the fans out with more filters, *or filters of lower resistance/larger media area* then you can realize the potential fully.
- I acknowledge that the best thing would be curves, lots of curves, but that's too much data to collect without more automation. So we just focus on the most relevant operating points, filter configurations etc.
- Spreadsheet with charts, same data (copied from this doc), some calculations, some info on cost etc.:
<https://docs.google.com/spreadsheets/d/1gbOp7BphLow2iQUmZIKjc041NLPL7MhYoHT93j9MLzs/edit?usp=sharing>
- Github with data files and all micropython and python programs:
<https://github.com/adouglas89/Big-quiet-air-purifier/>
- The particle sensor used was an SPS30 made by sensiron. The wiring diagram can be inferred from the micropython program, don't forget the pullup resistors for I2C You need the right connector for it too, see it's datasheet.
- The data is saved to an sd card, adapter module from adafruit, it has to be in smaller files rather than one big one due to ram limitations of the pico, then they are combined on the PC using a python program that is included in the github.

test protocol

initialization:

- Put mask on

Repeat for each test:

- Assemble system, set up instruments if haven't
 - Get mixing fans running and in place
 - Windows shut
 - Everything sealed.
 - Aerosol generator in place.
 - Test machine in place.
 - Get machine running with desired settings i.e. rpm. Measuring noise has to be done after or before the test, the mixing fans complicate things too much.
 - Thonny open with programs, recording and non recording pm sensor programs
 - Pico connected
 - Sd card in logging sensor device.
 - Get mask on
- Start aerosol release
- Name the folder in the python program
- Start pm sensor with non-logging program
- When it has reached about 100 (for the difference between pm1 and pm4, that is the contaminant we are measuring unless otherwise stated), halt aerosol production
- stop non logging program and start the logging program
- Start timer for 5 minutes, this is for the control test to measure background cadr of the room.
- Wait 5 minutes for control test to complete (diagnostic output in thonny will say >300000 msec since start, the alarm is just a reminder)
- After 5 minutes have elapsed since recording has started, turn on machine by plugging in power, do not adjust it in any way.
- Record
 - rpm of fan
 - Pressure when relevant, can be measured later with fan rpm as a reference point, for a given pair of fan/filter.
 - airflow velocity at relevant point using thermanemometer, whatever it says, adjust for calibration matrix later - minimal use during test. because the mixing fans cause stray currents. Can measure it another time at the same rpm etc when measuring noise.
 - We cannot measure noise at this point because the mixing fans produce too much background noise.
- Set timer for 6 minutes if expecting more than 500 cfm, 10 if less.
- Wait for timer to sound indicating test is complete
- Stop the program on the pico
- If desired, turn mixing fans off and log noise and velocity, also pressure, and velocity
- Turn the air purifier off

- Transfer data to pc using sd card, a working folder and also into a backup folder for all raw data so it never gets lost
- Copy analysis program into the working folder for the data
- Run the program to check things make sense, that test was valid. Don't do a bunch of tests without checking everything is right/working.
- Adjust the test point times using a straight edge of a piece of paper to find the lines of best fit for the data on the log-linear plot. Changing the test point times changes the slope on the log-linear plot of the assumed line that is used to calculate cadr (the line that connects the two points).
- Make notes and record the results in this document. Spreadsheet is for later it's hard to futz with.
- Then do next test.

Important notes:

- If the cadr is large compared to the volume of the test chamber, which is true, the result will tend to be biased in the downwards direction due to re-inhalation of the air that just recently passed through the machine, which will reduce total particle removal when compared to performance in a large room where the air exiting the device becomes mixed with the rest of the room before re-entering the machine.
- If the sensor is close to the machine, this could bias the apparent cadr upwards as some dirty air is sort of excluded? I'm not so sure about that. Anything that reduces net particle removal would reduce apparent cadr. Re-inhalation of air that is relatively clean would do that. However I'm not so sure about this phenomena.
- Mixing fans consist of the stand fan, 1 BQF at 400 rpm and 7x P14 Max fans at 12 volts. Combined maybe 3500-4000 cfm, and they entrain air at the exit so easily 6000 cfm total air motion in the room cross section.
- The room size is: 2544 cubic feet ($8.8333 \times 12 \times 24$). There is some stuff in the room, which consumes some volume. There are no closed closets etc.

test notes/data

Folder name: Bqap_10x_dafco_350_rpm_on_table_coscoveron:

Rpm 350 even though it says full power must have swapped the driver out at some point in the past for the lower voltage one.

Correct it later

control cadr: 223.10989679478965

test cadr before subtracting control: 976.2692585223523

apparent machine cadr: 753.1593617275627

Folder name: Bqap_10x_dafco_350_rpm_middle_of_room_coscoveron:

With the bqap in the middle of the room, or more like the western third

Mainly to compare with previous test which was same machine on a table at the end of the room, to determine what impact the location of the machine has on measured cadr.

Rpm is only 350 fuck why? I must have swapped out the driver with the lower voltage one. Well it's fine for this purpose

Air velocity during test at filter face : 0.33

control cadr: 63.304108878017274

test cadr before subtracting control: 874.9673265882021

apparent machine cadr: 811.6632177101849

Conclusion: the location of the machine relative to sensor and room mixing has a small impact on detected cadr. Also the background cadr varies considerably, probably due to wind.

Folder name: Bqap_10x_dafco_350_rpm_middle_of_room_NOcover:

Had to change folder name after test, forgot to set it before test.

Velocity: 0.38

control cadr: 138.51764308920082

test cadr before subtracting control: 998.4713067795774

apparent machine cadr: 859.9536636903765

Conclusion: the cover does have some impact on cadr but it's pretty small.

bqap_10x_dafco_full_power_middle_of_room_NOcoscoveron_401rpm

Bug in control box discovered, not actually run at 401 rpm but rather 350

control cadr: 193.57875539228505

test cadr before subtracting control: 945.7406700861397

apparent machine cadr: 752.1619146938547

What the fuck? The flow is clearly higher, and yet the cadr is lower? Unless the wind is fucking things up maybe, increasing control cadr, but even that does not really explain it as the figures before subtracting cadr are there. Fuck what the fuck? Obviously higher rpm must cause higher airflow and therefore higher cadr, not less. Leak in the machine? No. I did have to move it to reprogram the driver so it was capable of 400 rpm. Just try again and see what happens, if it's the wind it will be different.

Edit: the machine boots at 400 rpm but the control box has a voltage limit that I put in back when the fan needed it, which it no longer does. When the control box kicks in it reduces the rpm to 350. So a minor bug in the machine this explains the previous low rpm as well.

bqap_10x_dafco_full_power_middle_of_room_NOcoscoveron_401rpm_try2:

Failed first try, for some reason the machine reverted to 350 rpm! A minor bug in the control box. I unplugged the control box. The driver defaults to full power 400 rpm without it. This may explain the previous baffling results.

There is duplicate data from a previous failed test in the end of the file, that is why there are two sets towards the end. I simply changed the test point 2 to avoid that region.

In the future always make a new name for the folder, do not simply restart recording, to avoid this.

control cadr: 159.06062381897192

test cadr before subtracting control: 1185.2649781347084

apparent machine cadr: 1026.2043543157365

Pressure: 7.1 pascals

bqf_4x4x20x25_3Mfiltrete_full_power_middle_of_room_NOcoscover_408rpm:

The rpm was slightly higher due to the full power resistance to airflow being lower. The drive voltage, driver and fan are the same. This would have a very small but minimal effect on noise and should be measured. By turning the mixing fans off after the test is over and measuring it.

Noise measured with mixing fans off, at 1 meter to the side of the unit:

Background meter reading:36

Meter with machine on at 408 rpm:42.5

Anomalously low, this may be because the noise meter is on the nearby bed, try holding it in the air.

Reading with machine on and noise meter in my hand at 1 meter to the side: 43.9.

Background with noise meter in my hand in same location and machine off:34.6

control cadr: 140.76931370562883

test cadr before subtracting control: 1303.6194328303734

apparent machine cadr: 1162.8501191247447

Pressure: 10.8 pascals

Testo anemometer reading taken after turning mixing fans off to prevent stray motion: 0.89m/s

Discussion:

This is not much higher than the 300 rpm point according to the below, which is baffling. The data looks fine. I have no explanation really.

shit! forgot to cover the hole for the anemometer connector.

bqf_4x4x20x25_3Mfiltrete_full_power_middle_of_room_NOcoscover_300rpm:

“Full power but 300 rpm?” mistake in the naming but the test is valid. There are many more this error propagated fuck well teh data is fine and the notes are correct.

control cadr: 89.35892141208713

test cadr before subtracting control: 1154.6960570354522

apparent machine cadr: 1065.3371356233652 -> this is an outlier, exclude it from later analysis, we don't know what happened here but this does not appear to be right.

Pressure: 5.8 pascals

Velocity Testo anemometer reading taken after turning mixing fans off to prevent stray motion: 0.61 m/s

REDO this test. I did, below, like 7 times as I was testing the test method at the same time to determine standard deviation

bqap_10x_dafco_NOcoscoveron_300rpm:

Pressure: 4.315 pascals

Noise with mixing fans off: 39.5. The motor hum seems to have returned fuck.

Background noise: 36.5

How can the pressure be lower and yet the cadr is actually lower than with the 3m 4 inch filters?

Efficiency of the filters maybe?

control cadr: 94.93992867312035

test cadr before subtracting control: 745.8233692991365

apparent machine cadr: 650.8834406260162

Velocity: 0.3 m/s

Crap kind of low, lower than I thought oh well. The efficiency must go down with face velocity in some regions. With 1110 cfm flow measured previously, and 650 cfm cadr yeah that's pretty low efficiency, not up to the merv 13 spec for 1-4 micron particles.

I forgot to turn one of the mixing fans was not turned on. It's the smaller of the two, still a valid measure probably. This probably biassed the test downwards slightly though.

Tests to determine standard deviation of test method due to random noise:

bqf_4x4x20x25_3Mfiltrete_full_power_on_bed_300rpm:

RPM: 300

control cadr: 94.30643186951924

test cadr before subtracting control: 872.7045766136703

apparent machine cadr: 778.3981447441511

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-2:

control cadr: 94.14786451959515

test cadr before subtracting control: 943.2026362200835

apparent machine cadr: 849.0547717004883

The time of the data points had to be adjusted for better fit to avoid blips. Obviously this is a considerable part of the noise in this kind of data. I'm basically eyeballing a line of best fit, which is reasonable for a straight line. Might want to reprocess the data another day to use more sophisticated noise smoothing.

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-4:

Try 3 got messed up so we are skipping to number 4.

control cadr: 79.47466265161654

test cadr before subtracting control: 929.4653973447286

apparent machine cadr: 849.990734693112

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-5:

control cadr: 135.15228888026897

test cadr before subtracting control: 1008.7350574415418

apparent machine cadr: 873.5827685612728

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-6:

control cadr: 98.43813748582674

test cadr before subtracting control: 978.479831279529

apparent machine cadr: 880.0416937937023

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-7:

control cadr: 84.31694319124526

test cadr before subtracting control: 949.0031457062802

apparent machine cadr: 864.6862025150349

Again it's important to note I am eyeballing the line of best fit. I adjust the data point sample time slightly to get the line that they lie upon to be parallel to the eyeballed line of best fit. The exact points in time that are used are in the program, which is included with each data folder and are displayed graphically when the program is run.

bqf_4x4x20x25_3Mfiltrete_full_power_on_floor_mid_room_300rpm-8:

control cadr: 97.76200437125222

test cadr before subtracting control: 913.6108220252987

apparent machine cadr: 815.8488176540465

bqf_4x1inx20x25in_filtrete3m_300rpm:

control cadr: 175.2484430296277

test cadr before subtracting control: 964.0109847288843

apparent machine cadr: 788.7625416992566

Ok very high. Repeat this test later to be sure perhaps. However given the much higher pleating density maybe it's to be expected. The four inch filters have only 17 pleats. The 1 inch filters have 64 pleats. If it's the same filter media.

Bqf_4x1inx20x25in_filtrete3m_400rpm-2:

Pressure: 9.55 Pa

Airflow Velocity (middle of west facing filter while mixing fans are turned off, after the test.):0.72

control cadr: 233.00879907416245

test cadr before subtracting control: 1162.5344705545413

apparent machine cadr: 929.5256714803788

The (name)-1 test got borked, ran the wrong program for just a second when I meant to run the non logging program, which corrupted the data in the folder with that name, so that test was discarded.

The pressure indicates the flow is actually higher than it is for the 4 inch thick filters at this RPM. Lower pressure means higher flow is occurring through the fan at a given rpm.

Same filter and fan config but with the cover on, just a quick test to see what difference the cover makes, same 400 rpm:

velocity:0.6 m/s

pressure:10.55 Pa

So it does hit the flow but not that much.

Bqap_mpr1900_3mfiltrete_10x_1x20x25_400rpm-3:

First two got spoiled.

control cadr: 151.60479612415796

test cadr before subtracting control: 1097.5774497156958

apparent machine cadr: 945.9726535915379

pressure:7.15.

How can the pressure be lower, indicating higher flow, than the 4x box, same type of filters, yet nearly the same cadr?

Lower cadr than the dafco filters at same fan rpm, same fan???

Bqap_mpr1900_3mfiltrete_10x_1x20x25_300rpm-1:

Pressure: 4.35

control cadr: 188.79317744421604

test cadr before subtracting control: 810.7693687459242

apparent machine cadr: 621.9761913017082

Velocity: 0.35

Again, lower cadr than the same filters in 4x box form. I ran the test for 9 minutes this time to get more data and try to be sure. I have no explanation for this phenomena except maybe how air is moving around the intake of the fan is affecting things adversely. Or maybe lower face velocity is actually reducing efficiency.edit: it appears to be the low face velocity

Also notice the velocity is unusually low compared with the pressure when compared with the 4x box, indicating some constriction due to the internal geometry.

Bqf_merv13dafco_4x_1x20x25_300rpm-1:

control cadr: 165.02405213448955

test cadr before subtracting control: 647.8531747940609

apparent machine cadr: 482.82912265957134

Pressure: 7.35 Pa

Voltage: 15.8 V (lab supply)

Current: 0.78 A

Velocity: 0.4 m/s

Commentary: With the 10x dafco the cadr was nearly the same as using filtrete filters. With the 4x box the filtrete works much better. Oh well. That's not a surprise, the 3m ones are usually better.

Bqf_merv13dafco_4x_1x20x25_400rpm-1:

Pressure: 11.35 Pa

Voltage: 23.3 V lab power supply (the cheap one reads slightly low)

Current: 1.43

Rpm: 400

Velocity: 0.6 m/s

control cadr: 240.85605929161062

test cadr before subtracting control: 915.5554454049894

apparent machine cadr: 674.6993861133788

Why is the control cadr so high?? There is no wind or anything. It varies considerably.

Lasko_3733_dafco_4x_1x20x25_merv13_525rpm:

Noise reading of meter: 43.9 dBa

Background: 37 dBa

Voltage AC RMS: 100

Pressure: 3.87

Rpm: 525 rpm

Speed 1

control cadr: 161.58347312541676

test cadr before subtracting control: 460.1106285595063

apparent machine cadr: 298.52715543408954

Lasko_3733_dafco_4x_1x20x25_merv13_rpm_366rpm:

Noise reading: 38.5

Background noise: 34.8

Voltage rms ac: 74

Rpm: 366

Pressure: 2.2 Pa

Note: the hum of the motor seems to be making most of the noise

control cadr: 206.9214906204895

test cadr before subtracting control: 263.69576733461946

apparent machine cadr: 56.77427671412997

Note: remember the efficiency of the filters appears to go down as the velocity goes down.

Secondly this is approaching the noise floor, so this test shows only that the value is quite low, exactly what it is is relatively uncertain. The CADR seems disproportionately low compared with the pressure. Maybe run the test again

This one's an outlier, not clear what happened. Probably affected by wind during control period.

Lasko_3733_dafco_4x_1x20x25_merv13_rpm_382rpm:

Voltage: 78

Noise reading

Noise background:

Rpm 382

Pressure: not taken.

control cadr: 133.1394542237845

test cadr before subtracting control: 356.6302371763355

apparent machine cadr: 223.49078295255103

Note: I suspended testing when there was relatively much wind outside, and used the weather forecast to help predict calmer periods.

This one is useless the rpm is too low/below the noise value goal for some reason.

Lasko_3733_dafco_4x_1x20x25_merv13_rpm_447rpm:

Measured noise: 39.7

Background noise: 36.6

Rpm: 447

Voltage AC RMS: 92

control cadr: 176.75933574167627

test cadr before subtracting control: 441.90770696928706

apparent machine cadr: 265.1483712276108

Lasko_3733_filtrete_4x_1x20x25_mpr1900_449rpm:

Measured noise: 39.7

Background noise: 36.6

Rpm 449

Voltage ac rms: 86

control cadr: 156.22107270174533
test cadr before subtracting control: 458.13643990418655
apparent machine cadr: 301.9153672024412

Lasko_3733_filtrete_4x_1x20x25_mpr1900_bqAP_frame_plus20x20mpr1900onbottom_450rpm:

control cadr: 141.78973915987666
test cadr before subtracting control: 418.02709281943106
apparent machine cadr: 276.2373536595544

Same rpm same noise level presumably. How can it be a lower cadr when the bottom filter is added? A combination of lower filter efficiency due to lower velocity and error in the measurement perhaps. It would not be expected to make a big difference anyway.

bqf_filtrete_4x_1x20x25_mpr1900_bqAP_frame_plus20x20mpr1900onbottom_400rpm:

control cadr: 156.42918785441245
test cadr before subtracting control: 1192.1321127985902
apparent machine cadr: 1035.7029249441778

It sounds so much better than the lasko, no motor hum.

bqf_filtrete_4x_1x20x25_mpr1900_bqAP_frame_plus20x20mpr1900onbottom_400rpm_covers:

control cadr: 227.92273236769734
test cadr before subtracting control: 1084.915838632181
apparent machine cadr: 856.9931062644837

Hm so the cosmetic cover does have a significant impact unfortunately. More so than it does with the vvv option.

Lasko_3733_filtrete_4x_1x20x25_mpr1900_speed1:

Pressure: ?

Voltage: 120

Rpm:665

Pressure:4.37 Pa

Measured noise: 48.9 spl 1 meter from the edge of the machine. 1 meter from center:49.5.

Those measurements were made with the noise meter above the bed by 1 meter. If I measure on the other side of the machine, where there are hard background surfaces, it's 50.8 dBa.

Clearly nearby objects have an impact.

Background noise: 36.7

control cadr: 186.2548936006141

test cadr before subtracting control: 618.0370861405319

apparent machine cadr: 431.7821925399178

bqf_4x1x20x30_mpr220_3m_1x1x20x20_mpr1900_3m_400RPM:

control cadr: 101.54635000992407

test cadr before subtracting control: 962.4905801725301

apparent machine cadr: 860.9442301626061

Voltage: 24.3

Current: 1.6

Rpm: 400

Ok so that sucks, don't do that. Crap I had high hopes for that one. The legs may have a minor effect, they were 6 cm long

Lasko fan speed 3 4x1x20x25 filtrete mpr 1900 plus 1x20x20 mpr1900 filtrete on bottom (5 filter config):

Pressure:5.95 pascals.

Noise: 60.8 . background nearly irrelevant

Background noise:38 dBa (window open)

I guess I never got around to performing this test

BQF_6x_dafco_1x20x25_merv_13_300rpm:

300 rpm

control cadr: 139.72823076953804

test cadr before subtracting control: 705.5997930961275

apparent machine cadr: 565.8715623265895

This is with the single v in the middle, wooden frame, with the thicker blades fan, the thicker blades shouldn't make any difference.

BQF_6x_dafco_1x20x25_merv_13_400rpm:

400 rpm

control cadr: 58.96686540607321

test cadr before subtracting control: 955.0802848044497

apparent machine cadr: 896.1134193983764

Let's redo this test, the control cadr seems anomalously low, I forgot to turn on one of the mixing fans during the control phase. If leaving the fan off reduces background cadr, then the reduction x2 would be the error created. Because it is subtracted from the background cadr and then added to the test cadr.

BQF_6x_dafco_1x20x25_merv_13_400rpm-2:

control cadr: 85.9590369731942

test cadr before subtracting control: 899.5338379673203

apparent machine cadr: 813.5748009941261

Note: there was a mistake with this test, I forgot to tape up some gaps in the machine, this would have biased the result downwards. Since it is high, the filters are cheap and it might be the best combo, it would be a good idea to redo this test. Two in a v in the middle and four on the sides. That would eliminate the need for the 20 by 20 and also to use filtrete filters, if it goes well.

Bqf_4x1x20x25_dafco_merv-13_1x20x20_mpr1900_filtrete_nocover_400_rpm
wanted

Bqf_4x1x20x25_dafco_merv-13_1x20x20_mpr1900_filtrete_nocover_300_rpm
wanted

Bqf_4x1x20x25_dafco_merv-13_1x20x20_mpr1900_filtrete_withcoscover_400_rpm
Wanted

BQF_6x_filtrete_diagonalV_400rpm-1

Actually 403.8 rpm

Stepper fan turns out secondary was too far from primary

Small error on starting the machine for the first minute or so, the rpm was too low due to how the driver and the hysteresis works.

1.05 amps

24 volts

Should be 42.7 dba rating with 36 dBA background so 41.6 or so dba from the fan.

BQF_6x_filtrete_diagonalV_400rpm-2

Actually 404 rpm

Stepper fan, I think the coupler has been slowly coming off which causes the blades to be too far apart, reducing cadr. Adjust it. And try again for number 3. Measure the spacing.

It is making the occasional ticking sound apparently due to the blades falling off the shaft. yeah it's about 9 mm not the 3-4 it should be.

BQF_6x_filtrete_diagonalV_400rpm-3

400 rpm actually, gimbal motor small 69 mm one. *no mixing fan as I don't have one handy*.

Distance between secondary and primary when installed on frame is 6 mm.

control cadr: 142.0362230174004

test cadr before subtracting control: 1253.4419070815536

apparent machine cadr: 1111.4056840641533

Not really valid tests.

BQF_6x_filtrete_diagonalV_400rpm-4

This time with mixing fan, everything else the same.

Mixing fan bqf one turned off at 5 minute mark, let's see if it increases or decreases apparent cadr. Or no diff.

control cadr: 142.24496593296064

test cadr before subtracting control: 1052.5417794306602

apparent machine cadr: 910.2968134976995

Crap so for whatever reason the v in the middle doesn't seem to help any when using filtrete filters. May be the lower velocity leads to lower efficiency and thus lower cadr. Might be of benefit when using the crummiier filters. The CR foundation was also finding in the 1050 cfm cadr range. We could also look at other particle size ranges.

The more mixing appears to bias the cadr in the higher direction but it's not a major effect, mostly it's just more noise.

BQF_4x_20x25x1_filtrete_plus_column_400rpm-

Bqf mixing fan on sensorless driver on a lower speed doesn't make a huge diff and it's loud

control cadr: 119.11415532712621

test cadr before subtracting control: 1045.1439175740604

apparent machine cadr: 926.0297622469342

Sigh. so not as good as the single 20 inch on the bottom then. Should be measuring static pressure that's pretty inaccurate too though. Fan rpm with the right spacing is the best probably. Should probably redo that test too though.

BQF_4x_20x25x1_filtrete_plus_column_400rpm-2

Actually 405 rpm

control cadr: 161.84105930568114

test cadr before subtracting control: 927.7750848740792

apparent machine cadr: 765.934025568398

Wtf?

BQF_4x_20x25x1_filtrete_plus_column_400rpm-3

Actually 405 rpm

control cadr: 137.7416578090129

test cadr before subtracting control: 985.769044433608

apparent machine cadr: 848.027386624595

Ok fuck so that doesn't work. Probably the cheapie filters have higher resistance than the 20 inch filtrete or something. Maybe flow is slightly impeded

846 is the average of the above three tests.

todo

- Make an explanatory section of the document to describe how to analyze the data again should someone wish to do so.
- Go back and reprocess some of the data with a new program to select smaller particles, just check pm1.0 perhaps. Not combined with other stuff. That's basically smoke cadr. To see if the bqap works better or if another approach works better as face velocity could make a difference.

<https://www.amazon.ca/Lasko-3733-20-Fan-Box/dp/B00002ND67?th=1> has info on power of fan. I sent my power meter back, it might not work with the variac anyway. That says 53 on low, 93 watts on high.

Later:

- Backup all data